

# Dynamics of a Closed Low-Parameter Compartment Model of the Global Carbon Cycle

A. S. Ginzburg and N. N. Zavalishin

*A.M. Obukhov Institute of Atmospheric Physics, Russian Academy of Sciences, Pyzhevskii per. 3, Moscow, 119017 Russia*

*e-mail: gin@ifaran.ru*

*e-mail: nickolos@ifaran.ru*

Received December 12, 2007

**Abstract**—A closed dynamic compartment model of the global carbon cycle and its modifications with a small number of external parameters, which are constructed on the basis of data on the current and preindustrial carbon fluxes and reserves in reservoirs, are studied. The first possible controlling parameter is the amount of anthropogenic CO<sub>2</sub> emission into the atmosphere. Closure in substance allows a decrease in the dimensionality of dynamic models and yields another variable parameter—the system's total mass, whose change reflects the level of uncertainty in the knowledge of the amount of reserves. The calibration of unsaturated fluxes in each of the models is based on an algorithm developed earlier for local ecosystems, and the measured differences between the fluxes opposite in sign for several times are used to calculate the coefficients of saturated fluxes. The model's modifications make it possible to estimate the degree to which the oceanic active layer may be involved in rapid exchange with the atmosphere and to introduce the factor of land use as an additional anthropogenic effect. The time trajectories of carbon reserves in reservoirs are calculated until 2100 under variations in anthropogenic emissions in accordance with the IPCC basic scenarios until 2100 for the basic model and its modifications.

**DOI:** 10.1134/S0001433808060029

## INTRODUCTION: USING LOW-PARAMETER MODELS TO ANALYZE THE BEHAVIOR OF THE GLOBAL CARBON CYCLE

Studying the mechanisms of interaction between climate and the biosphere and predicting their evolution under anthropogenic forcing become more and more topical in relation to man's attempts to control certain important characteristics of this complex system under the Kyoto Protocol. Mathematical simulation is widely used for this purpose in view of the baffling complexity of the climate–biosphere system and significant uncertainty in the knowledge of its quantitative and qualitative characteristics on a global scale. The two most widespread approaches to stimulating this system are a spatially distributed approach, when the partial differential equations of climate theory are joined with a biospheric compartment in each of the land-surface cells, and a local point approach, in which the dynamics of carbon is described by a system of ordinary differential equations (ODEs) with parameters depending on climatic variables specified externally. There are also some versions in which these two approaches are combined. Most of the spatially distributed models of the global carbon cycle originate from the model described in SCOPE monographs [1] and the Moscow biospheric model [2]. Such models are represented by a large number of currently operating global models, among which there

are Russian models: V.F. Krapivin's model of global cycles [3], A.M. Tarko's model of carbon cycle in the atmosphere–vegetation–soil [4] and atmosphere–ocean systems, and a carbon compartment in the atmospheric general circulation model developed at the Institute of Numerical Mathematics of the Russian Academy of Sciences [5]. These models are characterized by different degrees of spatial resolution and specification of the basic biogeochemical processes that form the carbon cycle. As a rule, most of them use the global vegetation model (GVM) coupled with a cyclical model to estimate variations in the distribution of major biomes of the land during anthropogenic disturbances and climate changes [6].

In the models averaged over the globe, the cycle's biological components may be thought of as carbon reservoirs connected to one another by fluxes. Such models are referred to as zero-dimensional. Historically, the first of them was V.A. Kostitsyn's model (1935) [7]. A large number of such models were developed in the 1980s. The results of numerical experiments with the use of such models are described in [8, 9]. Models having four to seven compartments with both linear and nonlinear interactions are considered when an explicit time dependence of an anthropogenic CO<sub>2</sub> source is specified. A completely linear (in reserves) model demonstrates the only equilibrium state, which is reached at a rate dependent on the

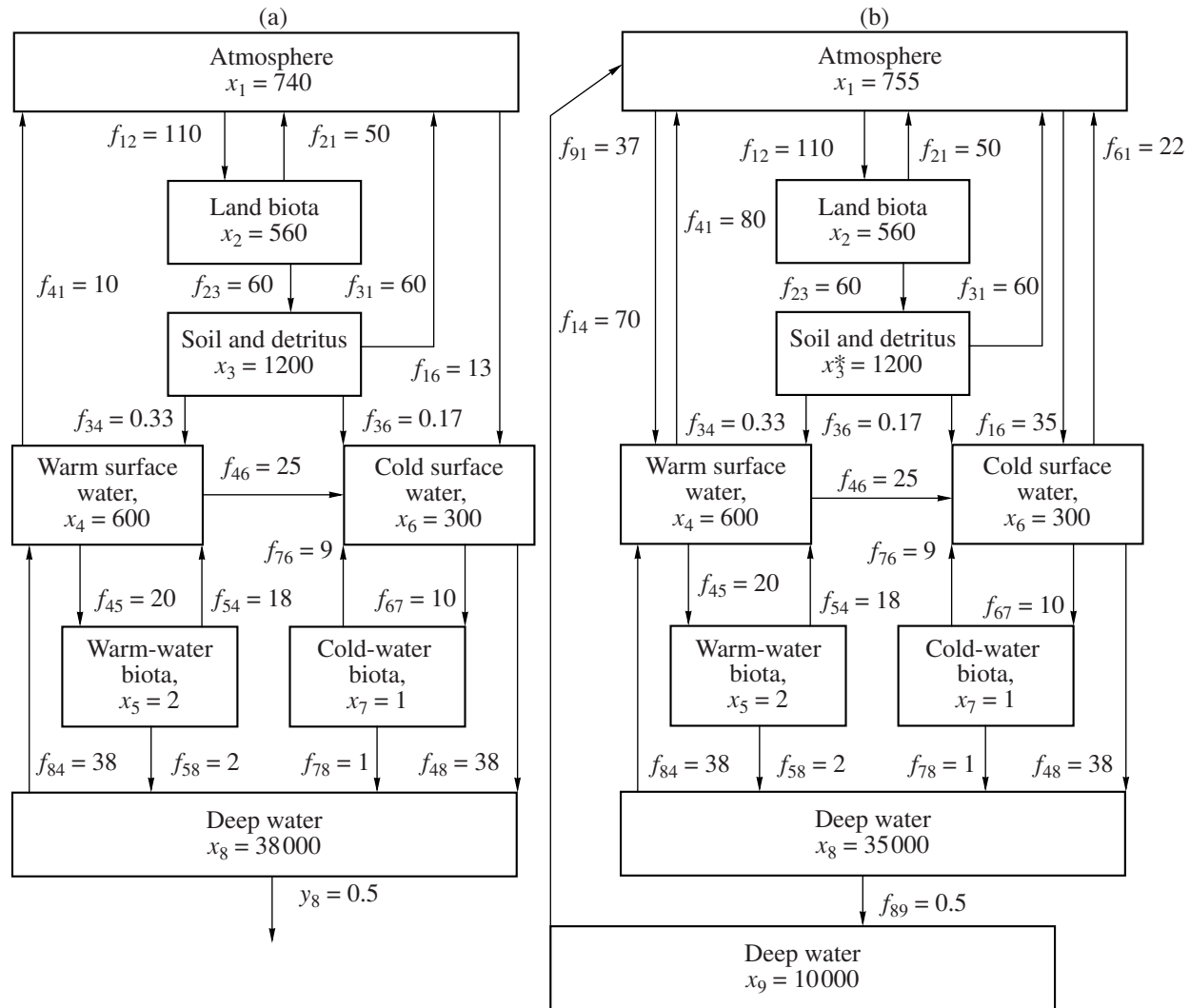
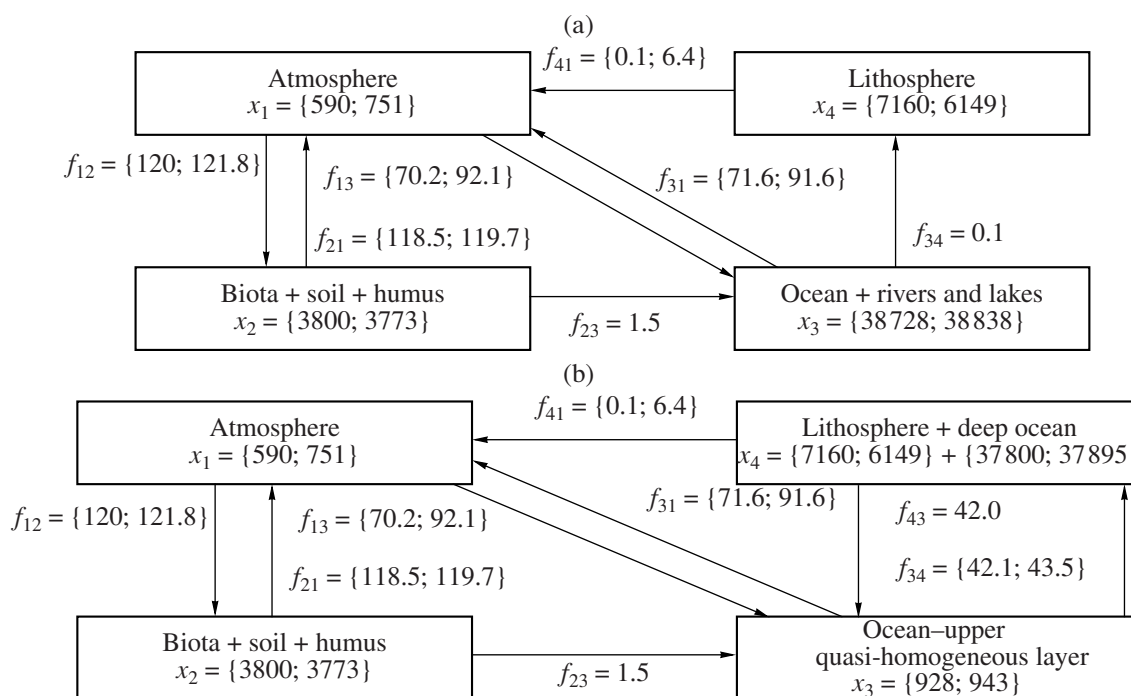


Fig. 1. Two schemes of the global carbon cycle according to (a) [13] and (b) [22]. Reserves in  $10^9$  tC, fluxes in  $10^9$  tC/year.

intensities of exchange fluxes between reservoirs. When there is a steady annual source of anthropogenic  $\text{CO}_2$ , the models of this type yield a linear increase in atmospheric carbon [8]. A new limiting state of the carbon cycle is characterized by transport of the carbon overweight to the deep ocean and by its elimination from the cycle's active phase. The same results are obtained with other compartment models with nonlinear forms of interactions between reservoirs, which differ in concrete predictions of their carbon content depending on their structure, the dynamics of industrial emissions, and land use. The models with the use of the annual mean land-surface temperature [10, 11] yield more diverse scenarios of carbon-cycle dynamics with multiple equilibria (virtual biospheres) and rapid (catastrophic) transitions between them. However, there have been almost no attempts to conduct a parametric analysis of these models. A qualitative or analytic study of even minimal zero-dimen-

sional models is the exception rather than the rule. Among such exceptions, one can mention the precursor of all models of global cycles, which was constructed by Kostitsyn [7]; Yu.M. Svirezhev and W. von Bloh's system of temperature-dependent cycles of carbon and water [10, 12]; and E.P. Borisenkov's linear nonautonomous model joined with the temperature equations of atmospheric layers [8, 9]. In Svirezhev–von Bloh's model, the structures of the phase portrait and the domains of attraction of different equilibrium states on the plane of the parameters corresponding to the total amounts of water and carbon on the globe are studied. However, the ocean is not used in this model, and this limits a prognostic value of the inferences about the dynamics of climate control by the biosphere under anthropogenic forcing, because the sensitivity of carbon exchange at the ocean–atmosphere boundary to climate changes has become the subject of active discussions in recent



**Fig. 2.** (a) Aggregated scheme of the global carbon cycle according to the SCOPE [20] and IPCC [15, 21] data. The values in braces correspond to the preindustrial and current states, respectively. Reserves (GtC) and their components:  $x_1$  corresponds to the atmosphere;  $x_2$  corresponds to the surface biota, soil, and humus;  $x_3$  corresponds to the ocean and the lake–river system (the upper photic layer + biota + deep ocean + lakes + rivers); and  $x_4$  corresponds to the exchange pool of the lithosphere. Fluxes (GtC/year):  $f_{12}$  corresponds to the absorption of  $\text{CO}_2$  by vegetation under photosynthesis (Gross Primary Production);  $f_{21}$  corresponds to vegetation and soil respiration, deforestation, and land use;  $f_{13}$  corresponds to the total absorption by the ocean and the lake–river system;  $f_{31}$  denotes carbon emissions by the ocean and the lake–river system;  $f_{23}$  corresponds to the removal of carbon from soil and its surface with river runoff to the ocean;  $f_{34}$  denotes precipitation on the ocean floor minus bottom-deposit dissolution; and  $f_{41}$  denotes volcanic emissions and fossil-fuel combustion. (b) Aggregated scheme of the global carbon cycle with the upper active layer of the ocean according to the SCOPE [20] and IPCC [15, 21] data. The values in brackets correspond to the preindustrial and current states, respectively. In the ocean, only the upper quasi-homogeneous layer is active, and the rest of the ocean and the lithosphere are in inertial reservoir 4. The fluxes  $f_{34}$  and  $f_{43}$  denote the exchange between the upper layer of the ocean and its deep waters.

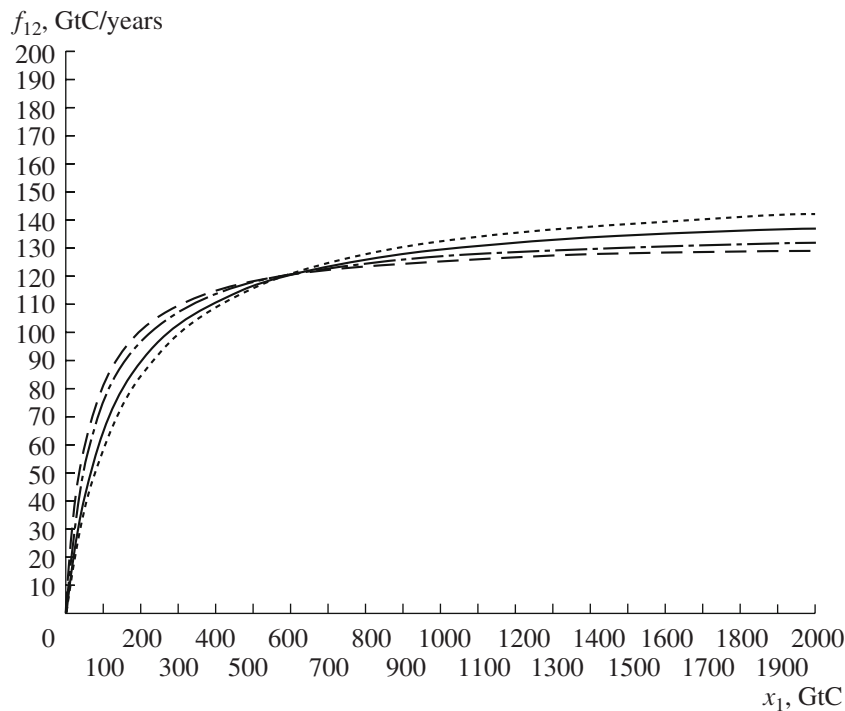
years [3]. O.-Y. Kwon and J.L. Schnoor [13] describe simulated numerical experiments with a compartment model developed on the basis of previously specified functional dependences of the specific fluxes of carbon per unit area or volume of the on the carbon concentration, with consideration for the mean global temperature of the surface air layer. The limitation of this approach is the necessity of considering the areas and volumes occupied by different compartments on a global scale, which is done in the model with the aid of constants, while these quantities can vary depending on climatic parameters. A similar model has been constructed by T.M. Lenton [14], which is characterized by parametrization of the fluxes of land photosynthesis and the ocean–atmosphere exchange. This model predicts the levels of carbon content in major reservoirs under the emission-control scenarios proposed in the IPCC third report [15]. On the basis of the Lenton and Kwon–Schnoor models, a carbon compartment of a climate model of intermediate complexity was constructed at the Institute of Atmospheric Physics, Russian Academy of Sciences (IAP RAS) [16, 17]. Other minimal models are either analyzed

numerically [18, 19] or demonstrate some isolated aspect of the problem without a sufficiently complete mathematical analysis. One more aspect of the problem is a certain uncertainty in the knowledge of many interactions between the processes that form the carbon cycle [3], and this makes it necessary to analyze a family of systems of equations with general properties rather than one rigidly specified system of equations.

The goal of this work is to study the stability and evolution of preindustrial, current, and potential dynamic regimes of the global carbon cycle during variations in climatic parameters and anthropogenic forcing on the basis of a zero-dimensional minimal compartment model and its modifications constructed from the diagrams of carbon reserves and fluxes.

#### AGGREGATION OF MULTIDIMENSIONAL SCHEMES OF THE GLOBAL CARBON CYCLE AND FORMS OF A DYNAMIC MODEL

The simulated scheme of the global carbon cycle is based on the collection of data on its current and pre-industrial states from the SCOPE book [20] and the



**Fig. 3.** Dependence of the flux  $f_{12}$  (absorption of atmospheric carbon by the surface biota) on atmospheric carbon mass. The basic coefficients are denoted by the solid curve according to formula (4), the dashed curve according to (3), and by the dotted and dot-and-dash curves according to the maximum and minimum values of the difference  $\Delta f_s$  (Table 1).

fourth report of the IPCC [21]. In some cases, estimates from the third report of the IPCC [15] are used. Additional information is taken from [13, 22], whose schemes were used earlier in [23] to simulate balances and are shown as an example of multicomponent schemes in Fig. 1.

Current knowledge of the carbon cycle contains information on atmosphere–land and atmosphere–ocean interactions and on the biosphere’s participation in carbon exchange; according to scientists’ views of the processes forming the carbon cycle and the relation of these processes to the Earth’s climate system, this information is of paramount importance. The data on carbon reserves and fluxes from [15, 20] have been aggregated in a four-compartment scheme containing the atmosphere ( $x_1$ ), the biota and soil ( $x_2$ ), the ocean with a lake–river system ( $x_3$ ), and the lithosphere ( $x_4$ ) as reservoirs. Below, this scheme is referred to as basic. Aggregation implies integration of several (probably dissimilar) compartments into one together with the corresponding fluxes. Figure 2a shows the numerical implementations of the basic scheme for the preindustrial and current states of the carbon cycle. One of the parameters of the system is the intensity of anthropogenic emissions, and the other is the total amount (in mass units) of carbon  $M$  in the reservoirs. Although the latter parameter is physically constant, some uncertainty in the knowledge of reserves in individual reservoirs makes it possible to consider this parameter varying and to eliminate the

variable, whose estimates and interactions have the greatest uncertainty, from the equations. The reserve of carbon in the lithosphere, which is estimated with a significant scatter and contains mainly the carbon that is excluded from the cycle for long periods due to precipitation processes in the ocean, can play the role of such a variable. The flux  $f_{41}$  is formed from two parts: natural emission caused mainly by a volcanic activity and anthropogenic flux as a result of industrial emissions. Therefore, let us represent  $f_{41}$  in the form of the linear function:  $f_{41} = q_{41} + \alpha_{41}x_4 = q_{41} + \alpha_{41}(M - \sum_{i=1}^3 x_i)$  with a natural emission of the donor type ( $\alpha_{41}x_4$  depends only on the reserve of a sending reservoir) and a constant value of the flux of anthropogenic emissions. Below, it will become clear that this flux can be represented only in the donor type without its constant part.

Since the atmosphere–land and atmosphere–ocean fluxes play the main role in the aggregate emission and absorption of carbon for time periods shorter than geological, the selection of their corresponding flux dependences is of critical importance for the model. A simple algorithm for selecting flux functions in the compartment models of ecosystems [24] has proven insufficient for this purpose, because it is necessary to take into account the effect of saturation of the  $\text{CO}_2$  flux according to the atmospheric content of carbon participating in vegetation photosynthesis. An expert knowledge of the mechanism of saturation in the process of  $\text{CO}_2$  absorption by land vegetation [3] shows

**Table 1.** Differences in the atmosphere–land fluxes and reserves in the atmosphere and surface biota [20, 21]

| Decades | $\Delta f_s = f_{12} - f_{21}$ | $x_1$ , GtC | $\Delta x_1$ , GtC/year | $x_2$ , GtC  | $\Delta x_2$ , GtC/year |
|---------|--------------------------------|-------------|-------------------------|--------------|-------------------------|
| 1980s   | $0.3 \pm 0.9$                  | $696 \pm 1$ | $3.3 \pm 0.1$           | $3786 \pm 9$ | $-0.3 \pm 0.9$          |
| 1990s   | $1.0 \pm 0.6$                  | $729 \pm 1$ | $3.2 \pm 0.1$           | $3783 \pm 6$ | $-1.0 \pm 0.6$          |
| 2000s   | $0.9 \pm 0.6$                  | 751         | $4.1 \pm 0.1$           | 3773         | $-0.9 \pm 0.6$          |

that this flux is saturated with an increase of atmospheric carbon content and corresponds functionally

to the Michaelis–Menton form  $f_{12} = \frac{K_{12}x_1\varphi(x_2)}{L_{12} + x_1}$ ,

which requires calculation of two parameters for its complete determination—the semisaturation value  $L_{12}$  and the fertilization constant  $K_{12}$ . Such a representation of the photosynthetic flux of vegetation with some variations is characteristic of most global carbon models [13, 16, 25]. The function  $\varphi(x_2)$  reflects the influence of the carbon content in vegetation and soil on  $\text{CO}_2$  absorption, and it is usually assumed to be linear ( $\varphi = x_2$ ). However, the physiology of plants and the knowledge of their competition for light and mineral resources suggest a nonlinear character of this dependence. At an increased content of carbon in vegetation, photosynthesis can undergo not only saturation, it can also be suppressed; this is manifested in certain global models of the carbon cycle [7]. In the Svirezhev–von Bloh [10, 12] and King (see [25]) models, this effect is expressed as a logistic dependence of primary productivity on the content of carbon in vegetation. As a rule, the function  $\varphi$  has either

the hyperbolic form  $\varphi = \frac{x_2}{N_{12} + x_2}$ , where  $N_{12}$  is the

constant of flux semisaturation in surface-biota carbon, or the conventional logistic form  $\varphi = x_2(M - x_2)$ .

The backward flux from the land to the atmosphere is formed from the combined respiration of vegetation and soil, which can be taken linearly proportional to the reserve of carbon in reservoir 2 (donor type):  $f_{21} = \alpha_{21}x_2$ . The land-use emission can be taken into account with the aid of the summand  $q_{21}$ , which is time-dependent and added to the linear function, or the variable coefficient  $\alpha_{21}$ .

The carbon exchange at the atmosphere–ocean boundary is complex in character because it is determined by the structure of the carbonate system of the ocean [4]. As a rule, in zero-dimensional models, the flux of the donor type  $f_{13}$  is used, because it is proportional to the partial pressure of carbon dioxide in the

atmosphere:  $f_{13} = \alpha_{13}x_1$  [3]. The form of the opposite flux does not keep within a simple scheme of selecting flux functions in compartment models; however, as a rough approximation, the flux can be represented in the Volterra form  $f_{31} = \gamma_{31}x_1x_3$ , because equilibrium in the carbonate system is established rapidly [4]. This implies that, if the concentration of atmospheric carbon increases slightly, its absorption by the ocean increases as well as the ocean response that ensures the recovery of balance. The precipitation of carbon from the water mass of the ocean to its floor is described by the interaction of the donor type:  $f_{34} = \alpha_{34}x_3$  [13].

The described selection of the flux functions results in a system of four ordinary differential equations of the basic model, whose dimensionality can be decreased by one owing to the constancy of the total carbon mass  $\sum_{i=1}^4 x_i = M = \text{const}$ :

$$\begin{cases} \frac{dx_1}{dt} = q_{41} + \alpha_{41}(M - x_1 - x_2 - x_3) \\ \quad - \frac{K_{12}x_1\varphi(x_2)}{L_{12} + x_1} + \alpha_{21}x_2 - \alpha_{13}x_1 + \gamma_{31}x_1x_3, \\ \frac{dx_2}{dt} = \frac{K_{12}x_1\varphi(x_2)}{L_{12} + x_1} - \alpha_{21}x_2 - \alpha_{23}x_2, \\ \frac{dx_3}{dt} = \alpha_{13}x_1 - \gamma_{31}x_1x_3 + \alpha_{23}x_2 - \alpha_{34}x_3. \end{cases} \quad (1)$$

At  $q_{41} = 0$ , we obtain the system corresponding to the preindustrial state.

It follows from geochemical data that not all the carbon of the world ocean participates in the global cycle with the same rate of circulation. Therefore, the four-compartment scheme under consideration can easily be transformed into a scheme with a thin upper layer of the ocean without an increase in the number of variables. In this modification (Fig. 2b), the reservoir with the greatest uncertainty in reserve values contains (in addition to the lithosphere) carbon of a



**Table 2.** Variability of the coefficients of the fluxes  $f_{12}$  and  $f_{21}$  in models (1) and (1-M)

| Coefficients of the fluxes $f_{12}$ and $f_{21}$<br>at $\Delta f_s = [0.3 - 0.9; 0.3; 0.3 + 0.9]$ |          |                    |                    |
|---------------------------------------------------------------------------------------------------|----------|--------------------|--------------------|
| $L_{12}$                                                                                          | $K_{12}$ | $\alpha_{21}^{90}$ | $\alpha_{21}^{00}$ |
| 168.06                                                                                            | 0.040574 | 0.032562           | 0.032730           |
| 86.129                                                                                            | 0.036189 | 0.032707           | 0.032916           |
| 125.06                                                                                            | 0.038272 | 0.031940           | 0.032068           |

deep-water layer of the ocean, and the ocean–atmosphere exchange occurs only owing to its upper active layer. This upper quasi-homogeneous layer of the ocean exchanges carbon with its deep-water layer through the fluxes  $f_{34}$  and  $f_{43}$  of the donor type of control:  $f_{34} = \alpha_{34}x_3$  and  $f_{43} = \alpha_{43}(M - x_1 - x_2 - x_3)$ . Thus, the system of dynamic cycle equations may be written as

$$\begin{cases} \frac{dx_1}{dt} = q_{41} + \alpha_{41}(M - x_1 - x_2 - x_3) \\ \quad - \frac{K_{12}x_1\varphi(x_2)}{L_{12} + x_1} + \alpha_{21}x_2 - \alpha_{13}x_1 + \gamma_{31}x_1x_3, \\ \frac{dx_2}{dt} = \frac{K_{12}x_1\varphi(x_2)}{L_{12} + x_1} - \alpha_{21}x_2 - \alpha_{23}x_2, \\ \frac{dx_3}{dt} = \alpha_{13}x_1 - \gamma_{31}x_1x_3 + \alpha_{23}x_2 - \alpha_{34}x_3 \\ \quad + \alpha_{43}(M - x_1 - x_2 - x_3). \end{cases} \quad (1-M)$$

### UNIFIED APPROACH TO CALIBRATION OF THE FLUX FUNCTIONS

The unified approach to calibrating the coefficients of model (1) is that the coefficients of simple one-parameter fluxes are calculated directly from the values of reserves and fluxes for a selected scheme (for example, the current scheme) [24], and the coefficients of more complex dependences of the  $f_{12}$  type can be found from the flux differences measured for several times. The universality of this approach lies in the fact that the number of calibrated flux coefficients can be chosen according to the amount of available data. The notion of a complete equilibration of carbon fluxes in nature before the start of man's significant activities—the equilibrium preindustrial state  $x^*$  [8] dated to 1850—is traditional for globally averaged models. Since the flux caused by anthropogenic emissions did not exist in the preindustrial carbon cycle, it is necessary to consider different dynamic systems, one of which corresponds to the preindustrial equilibrium and the others correspond to the states at times close to the current time. The flux that corresponds to land use and contains the carbon losses from soil due to plowing and deforestation existed before 1850; however, according to estimates from [14], it was small compared to natural fluxes; therefore, it can be treated as a small perturbation of a steady state.

The parametrization of the flux  $f_{12}$  according to the preindustrial ( $x^*$ ,  $F^*$ ) and current (the 1990s;  $x^0$ ,  $F^0$ ) compartment schemes (Fig. 2) proves to be insufficiently adequate. If the second equilibrium equation from (1), where the coefficients  $\alpha_{21}^*$  and  $\alpha_{23}^*$  are cal-

**Table 3.** Coefficients of unsaturated fluxes for low-parameter models (1) and (1-M)

| Flux coefficients          | System (1) in equilibrium according to the IPCC (2001) and SCOPE (2004) | System (1) for current conditions according to IPCC (2001) and SCOPE (2004) | System (1-M) in equilibrium according to the IPCC (2001) and SCOPE (2004) | System (1-M) for current conditions according to the IPCC (2001) and SCOPE (2004) |
|----------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $\alpha_{41}$              | $1.3966 \times 10^{-5}$                                                 | $1.0411 \times 10^{-3}$                                                     | $2.2705 \times 10^{-6}$                                                   | $1.458 \times 10^{-4}$                                                            |
| $\alpha_{13}, \alpha_{31}$ | 0.11898                                                                 | 0.12264                                                                     | $7.7155 \times 10^{-2}$                                                   | $9.7137 \times 10^{-2}$                                                           |
| $\alpha_{21}$              | $3.1184 \times 10^{-2}$                                                 | $3.1725 \times 10^{-2}$                                                     | $3.1184 \times 10^{-2}$                                                   | $3.1725 \times 10^{-2}$                                                           |
| $\alpha_{23}$              | $3.9474 \times 10^{-4}$                                                 | $3.9756 \times 10^{-4}$                                                     | $3.9474 \times 10^{-4}$                                                   | $3.9756 \times 10^{-4}$                                                           |
| $\alpha_{34}$              | $2.5821 \times 10^{-6}$                                                 | $2.5748 \times 10^{-6}$                                                     | $4.5366 \times 10^{-2}$                                                   | $4.6129 \times 10^{-2}$                                                           |
| $\gamma_{31}, \gamma_{13}$ | $3.1335 \times 10^{-6}$                                                 | $3.1405 \times 10^{-6}$                                                     | $1.2821 \times 10^{-4}$                                                   | $1.3005 \times 10^{-4}$                                                           |
| $\alpha_{43}$              | —                                                                       | —                                                                           | $9.5359 \times 10^{-4}$                                                   | $9.5683 \times 10^{-4}$                                                           |

culated in equilibrium, is taken into account, the coefficients  $K_{12}$  and  $L_{12}$  can be determined from the formulas

$$L_{12} = \frac{x_1^* x_2^0 (f_{12}^0 x_2^* - (\alpha_{21}^* + \alpha_{23}^*) x_2^0)}{(\alpha_{21}^* + \alpha_{23}^*) x_2^0 x_1^0 - f_{12}^0 x_2^* x_1^*} = 66.439,$$

$$K_{12} = \frac{f_{12}^0}{x_2^0} \left( 1 + \frac{x_1^* (f_{12}^0 x_2^* - (\alpha_{21}^* + \alpha_{23}^*) x_2^0)}{(\alpha_{21}^* + \alpha_{23}^*) x_2^0 x_1^0 - f_{12}^0 x_2^* x_1^*} \right) \quad (2)$$

$$= 0.035138,$$

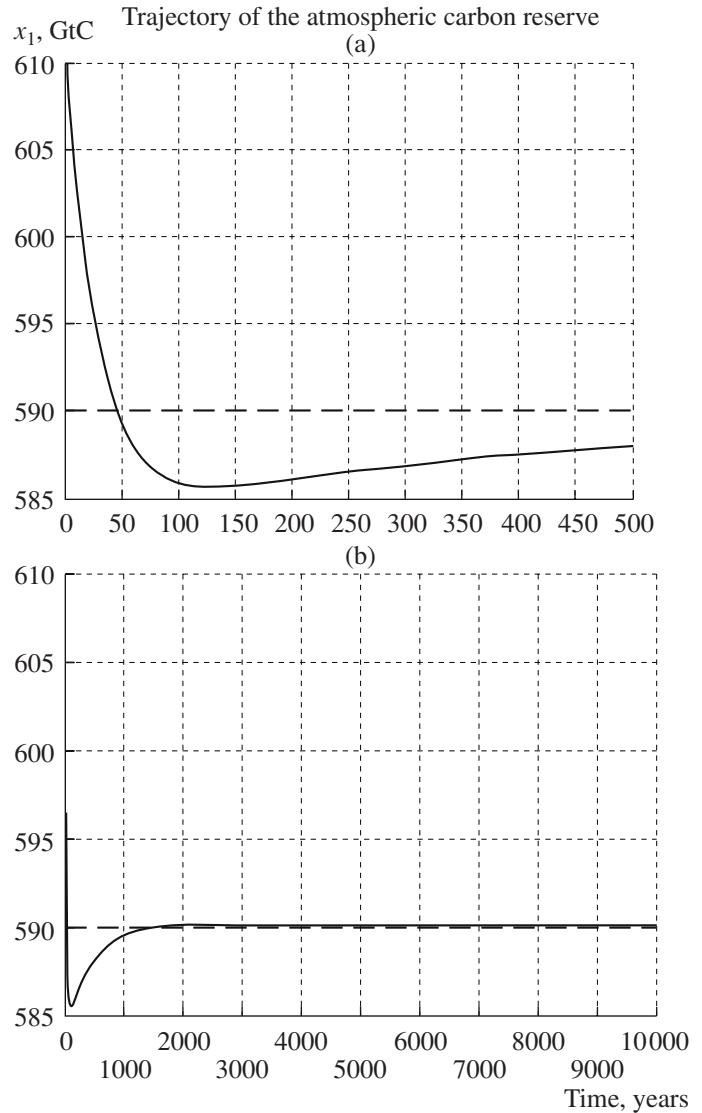
which yield a very low value of flux saturation, as if the level of total saturation had already been reached at present (Fig. 3).

A more correct calibration of coefficients is obtained using the difference between the fluxes  $\Delta f_s = f_{12} - f_{21}$  estimated for several times in recent decades and indicated in the IPCC report [21]. This zero difference is formed from the flux corresponding to land use and entering into  $f_{21}$  and additional vegetation absorption (residual sink) due to fertilization, which is included in  $f_{12}$ . To calibrate the coefficients  $K_{12}$ ,  $L_{12}$ , and  $\alpha_{21}$ , we can use the data on the nonzero differences  $\Delta f_s$  for the 1980s, 1990s, and 2000s, which are given in Table 1 together with the level of measurement uncertainty. Average (over these decades) variations in the content of atmospheric carbon are specified in the IPCC report [21], and the reserves in the surface biota for the 1980s and 1990s are estimated from their mean annual variations for both decades. If  $K_{12}$ ,  $L_{12}$ , and  $\alpha_{21}$  for the 2000s are believed to be unknown, for their determination, we have the system of algebraic equations

$$\begin{cases} \Delta f_s^{80} = \frac{K_{12} x_1^{80} x_2^{80}}{L_{12} + x_1^{80}} - \alpha_{21}^{80} x_2^{80}; \\ \Delta f_s^{90} = \frac{K_{12} x_1^{90} x_2^{90}}{L_{12} + x_1^{90}} - \alpha_{21}^{90} x_2^{90}; \\ \Delta f_s^{00} = \frac{K_{12} x_1^{00} x_2^{00}}{L_{12} + x_1^{00}} - \alpha_{21}^{00} x_2^{00}; \\ f_{12}^* = \frac{K_{12} x_1^* x_2^*}{L_{12} + x_1^*}, \end{cases} \quad (3)$$

which consists of three one-type relations for differences and the equation for the flux  $f_{12}$  under the conditions of preindustrial equilibrium and is unambiguously solved:

$$L_{12} = \frac{x_1^* x_1^{80} [x_2^{80} (\Delta f_s^{80} + \alpha_{21}^{80} x_2^{80}) - f_{12}^* x_2^{80}]}{f_{12}^* x_1^{80} x_2^{80} - x_1^* x_2^* (\Delta f_s^{80} + \alpha_{21}^{80} x_2^{80})},$$



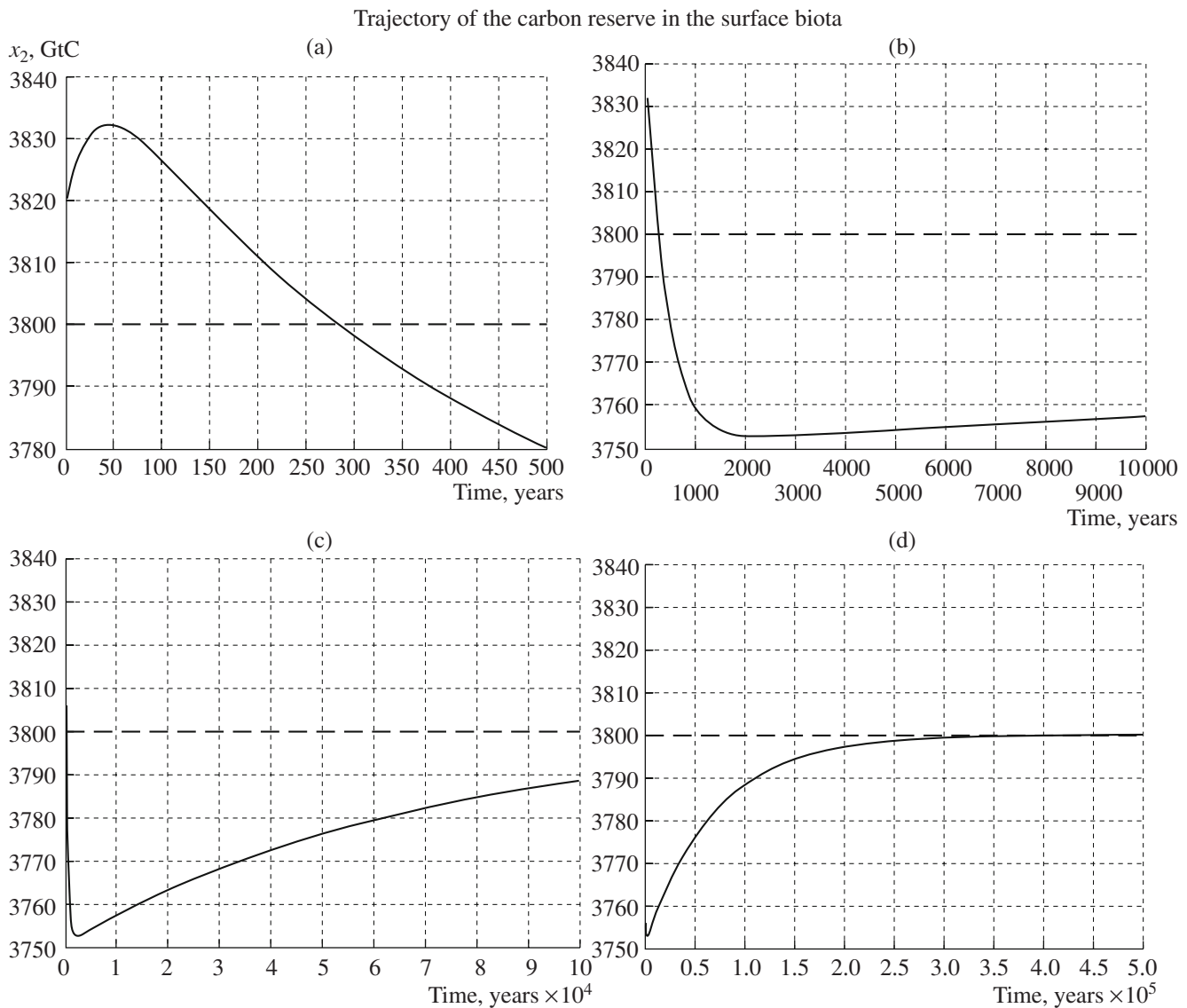
**Fig. 4.** Reserves of atmospheric carbon (GtC) over (a) 500 and (b) 10 000 years after a single disturbance of the preindustrial equilibrium in model (1).

$$K_{12} = \frac{f_{12}^* (L_{12} + x_1^*)}{x_1^* x_2^*},$$

$$\alpha_{21}^{80} = \frac{1}{x_2^{80}} \left( \frac{K_{12} x_1^{80} x_2^{80}}{L_{12} + x_1^{80}} - \Delta f_s^{80} \right),$$

$$\alpha_{21}^{90} = \frac{1}{x_2^{90}} \left( \frac{K_{12} x_1^{90} x_2^{90}}{L_{12} + x_1^{90}} - \Delta f_s^{90} \right). \quad (4)$$

The choice of  $\alpha_{21}^{00}$  as an unknown is due to the fact that, up to now, there have been no reliable estimates for the carbon cycle corresponding to land use and additional ecosystem absorption for 2000–2005 [20, 21]. The variation of the total emission of carbon from the sur-



**Fig. 5.** Carbon reserve (GtC) in the surface biota over (a) 500, (b) 10000, (c) 100000, and (d) 500000 years after a single disturbance of the preindustrial equilibrium in model (1).

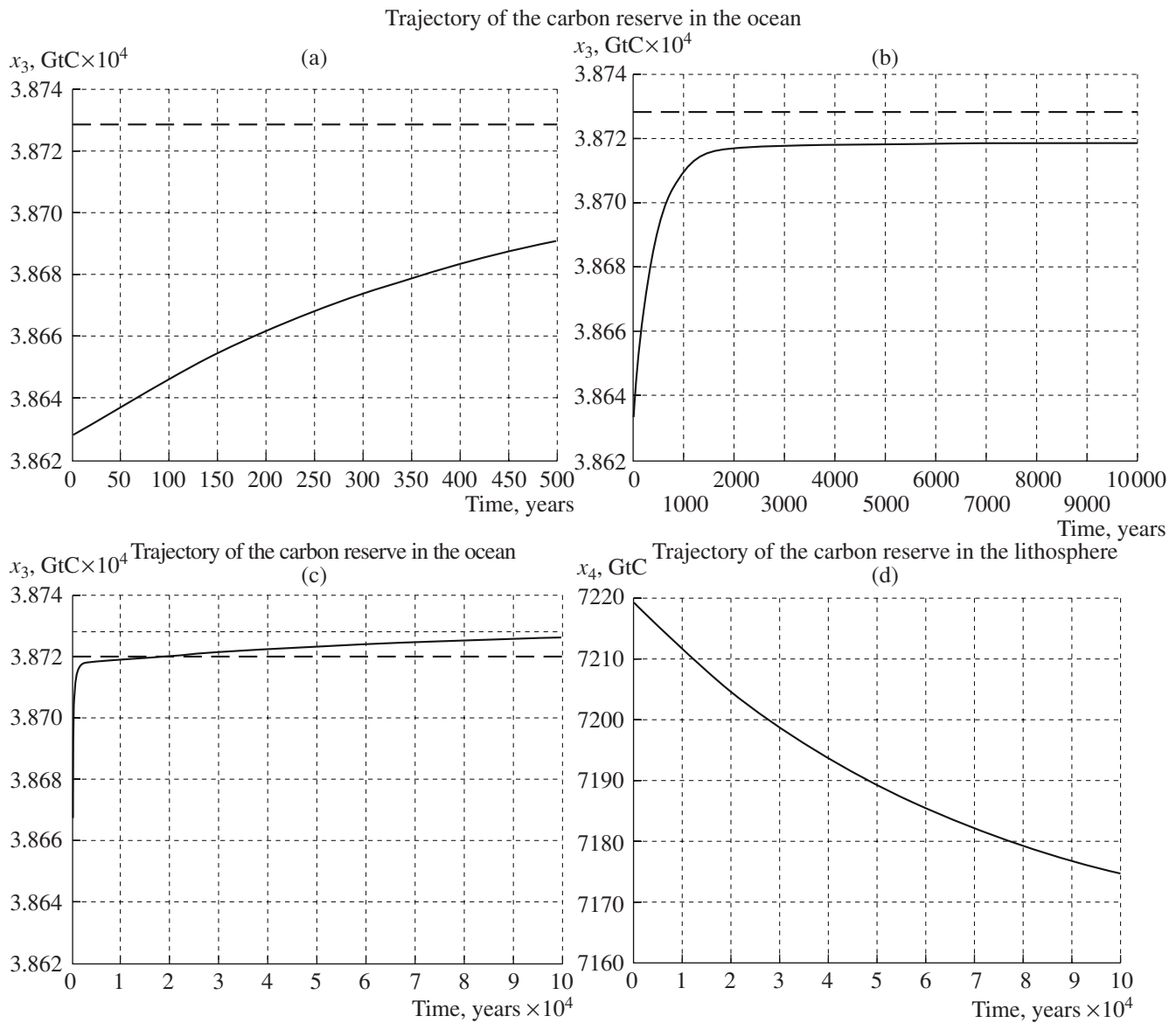
face biota due to the anthropogenic flux of land use, thus, enters into the coefficient  $\alpha_{21}$ . Table 2 gives the coefficient values (calculated from formulas (4)) under variations in  $\Delta f_s$  in the 1980s, and Fig. 3 shows the range of the functional dependences of surface-vegetation photosynthesis on the content of atmospheric carbon. It follows from Table 2 that the semisaturation constants  $L_{12}$  for the lower and upper limits of  $\Delta f_s$  differ by a factor of two, and the intensity of the total respiration of the surface biota varies only slightly. Figure 3 shows that the calculations with the use of the known differences yield more adequate graphs than those obtained from the calculations based only on two schemes of equilibrium and current states, even in spite of the fact that the surface biota in the four-reservoir

scheme consists of the vegetation and soil reservoirs, which function differently. The results of calculation of the intensities of the other fluxes in systems (1) and (1-M) for the equilibrium and current states are given in Table 3.

#### BASIC MODEL: PROPERTIES AND DYNAMICS

The nonnegative cone of family (1) at  $q_{41} = 0$  is invariant with respect to the effect of phase flux, because the right-hand side of each of the equations is nonnegative on the hyperplane that corresponds to the zero coordinate. If  $q_{41} > 0$ , it is necessary to consider the equation for  $x_4$  to make certain that the invariance





**Fig. 6.** Carbon reserve (GtC) in the ocean over (a) 500, (b) 10000, (c) 100000, and (d) 500000 years after a single disturbance of the preindustrial equilibrium in model (1).

is lost:  $\frac{dx_4}{dt} = -q_{41} - \alpha_{41}x_4 + \alpha_{34}x_3 < 0$  at  $x_4 = 0$ ; therefore, the phase trajectory leaves the positive orthant. At the same time, it is possible to show that system (1) is dissipative, i.e., as time passes, any trajectory finds itself inside a sphere with a finite radius in the phase space, and this indicates that the model is adequate. Indeed, let  $w = \sum_{i=1}^{n-1} x_i$ , then the following inequality is valid:

$$\frac{dw}{dt} + \alpha_{41}w < q_{41} + \alpha_{41}M - \alpha_{34}x_3 < q_{41} + \alpha_{41}M.$$

The inhomogeneous linear equation corresponding to this inequality has a solution of the form:

$$w = \left( \frac{q_{41}}{\alpha_{41}} + M \right) (1 + Ce^{-\alpha_{41}t}) < \frac{q_{41}}{\alpha_{41}} + M = \text{const},$$

which provides an asymptotic limitation of the trajectories of the reduced system. Hence, it follows that anthropogenic (as well as natural) emissions can be represented as a linear function of the donor type with the variable coefficient  $\alpha_{41}$ ; in this case, reservoir  $x_4$  loses carbon exponentially but not linearly as in the presence of a constant. Therefore, below, we will consider system (1) without the summand  $q_{41}$  and

keep in mind that the parameter  $\alpha_{41}$  varies from  $1.3966 \times 10^{-5}$  or higher with its current value 0.0010411 that corresponds to an amount of anthropogenic emissions of 7.2 GtC for 2005. The same is true for model (1-M).

System (1) has a pair of equilibria of the form  $x^{(1)} = [x_1^{(1)} \ 0 \ x_1^{(3)}]$ , which are referred to as atmospheric–oceanic equilibria, and a nontrivial equilibrium with zero components  $x^{(2)}$ , which is calculated from the formulas

$$\begin{aligned} x_1^{(2)} &= \frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}}, \\ x_3^{(2)} &= \frac{\alpha_{41}[M\alpha_{23}(K_{12} - \alpha_{21} - \alpha_{23}) - (\alpha_{23} - \alpha_{13})L_{12}(\alpha_{23} + \alpha_{21})]}{\alpha_{34}(\alpha_{23} + \alpha_{41})(K_{12} - \alpha_{21} - \alpha_{23}) + \alpha_{41}[\alpha_{23}(K_{12} - \alpha_{21} - \alpha_{23}) + \gamma_{31}L_{12}(\alpha_{23} + \alpha_{21})]}, \\ x_2^{(2)} &= \frac{1}{\alpha_{23}} \left[ x_3^{(2)} \left( \alpha_{34} + \gamma_{31} \frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}} \right) - \alpha_{13} \frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}} \right]. \end{aligned} \quad (5)$$

For the preindustrial state, the latter is a stable node, because the eigenvalues of its Jacobi matrix are  $-0.0326$ ,  $-1.4 \times 10^{-5}$ , and  $-0.00245$ . If anthropogenic carbon stops coming into the atmosphere from the lithosphere, the relaxation times for this state in different reservoirs differ markedly. In the atmosphere, in the case of a single disturbance, the preindustrial equilibrium content of carbon is recovered approximately during 1200 years (Fig. 5a). After that, the redistribution of excess carbon occurs only between the surface biota and the ocean, which need 250000 and 150000 years (Figs. 5, 6), respectively, and the lithospheric reserve, which, in this case, symbolizes bottom sediments and

volcanic carbon, is recovered even more slowly (Fig. 6d) in view of the smallness of natural fluxes of precipitation on the ocean floor and volcanic emanations. However, the process of recovery is almost linear and has a quasi-stationary character for times on the order of decades (Figs. 4a, 5a, 6a). Note that, in the absence of anthropogenic emissions, the preindustrial equilibrium is also established in the case when the system starts up with the initial conditions corresponding to the current situation.

The Jacobi matrix of system (1) for the nontrivial equilibrium has the form

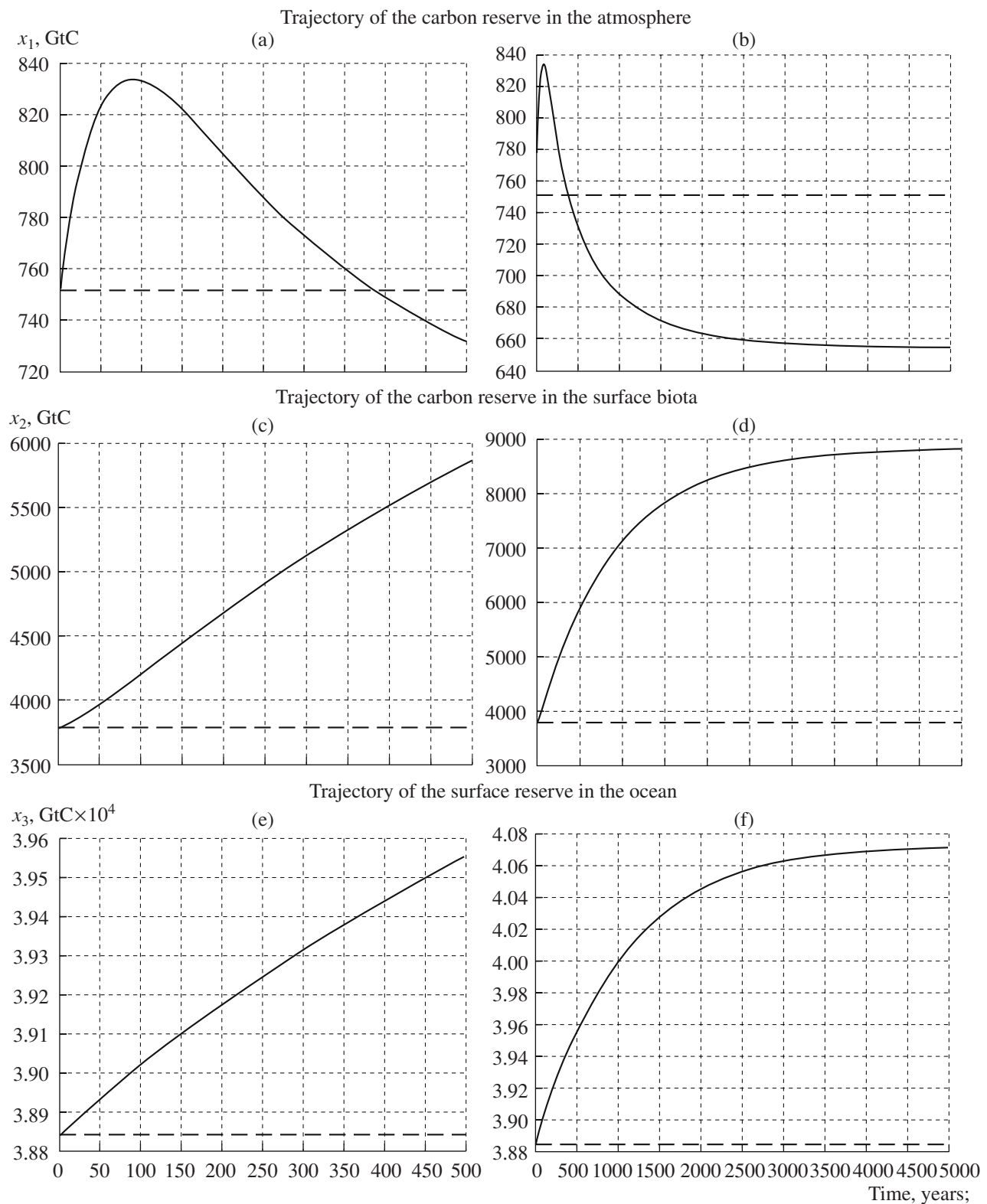
$$J^{(2)} = \begin{bmatrix} a & b & c \\ d & 0 & 0 \\ f & g & h \end{bmatrix} = \begin{bmatrix} -\alpha_{41} - \alpha_{13} - \frac{K_{12}L_{12}x_2^{(2)}}{(L_{12} + x_1^{(2)})^2} + \gamma_{31}x_3^{(2)} - \alpha_{41} - \frac{K_{12}x_1^{(2)}}{L_{12} + x_1^{(2)}} + \alpha_{21} & \gamma_{31}x_1^{(2)} - \alpha_{41} \\ \frac{K_{12}L_{12}x_2^{(2)}}{(L_{12} + x_1^{(2)})^2} & 0 & 0 \\ \alpha_{13} - \gamma_{31}x_3^{(2)} & \alpha_{23} & -\gamma_{31}x_1^{(2)} - \alpha_{34} \end{bmatrix}, \quad (6)$$

and its characteristic equation  $\lambda^3 + k_2\lambda^2 + k_1\lambda + k_0 = 0$  with the coefficients is expressed through the system's parameters:  $k_2 = -(a + h)$ ,  $k_1 = -(bd + cf - ah)$  and  $k_0 = d(bh - cg)$ . The line corresponding to the equality  $k_0 = d(bh - cg) = 0$  on the plane of the parameters  $(\alpha_{41}, M)$  corresponds to the loss of stability of a stationary solution by means of intersection or merging with one of the unstable atmospheric–equilibria,

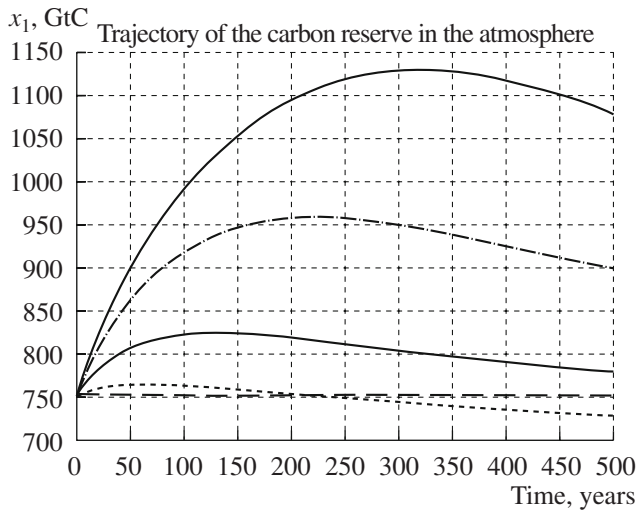
$$M = \frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}} + \frac{\alpha_{13}(\alpha_{34} + \alpha_{41})}{\alpha_{41}(\alpha_{34}(K_{12} - \alpha_{21} - \alpha_{23}) + \gamma_{31}L_{12}(\alpha_{21} + \alpha_{23}))} =$$

$9459.7 + \frac{0.022701}{\alpha_{41}}$  at the current parameters of the carbon cycle. The preindustrial equilibrium loses stability below the graph of this function, i.e., at very small values of the total mass of carbon in the system ( $M < 10000$  GtC). The line of generation of self-oscillations is expressed from the equality  $R_2 = k_2k_1 - k_0 = -(a + h)(ah - cf) + d(ab - cg)$  and is not present in the positive orthant of the parametric plane.

The dynamics of carbon reserves in the atmosphere, surface biota, and ocean at a continuous anthropogenic flux with the intensity of the 1990s is shown in Fig. 7, where the year 1999 is taken as a zero time. The model shows that the content of atmo-



**Fig. 7.** Carbon reserves (GtC) in the (a, b) atmosphere, (c, d) surface biota, and (e, f) ocean over 500 and 2000 years under the influence of constant-intensity anthropogenic emissions with a flux of 7.2 GtC/year. The dashed line corresponds to the reserve values for the early 2000s.



**Fig. 8.** Dynamics of atmospheric carbon under the influence of constant-intensity anthropogenic emissions of a with a flux of 7.2 GtC/year in model (1) for different values of the photosynthesis coefficients  $K_{12}$  and  $L_{12}$ . The basic coefficients are denoted by the solid line according to formula (4), the dashed line according to (3), and the dotted and dot-and-dash lines according to the maximum and minimum values of the difference  $\Delta f_3$  (Table 1), respectively.

spheric carbon increases over the course of 100 years to 830–850 GtC (Fig. 7a) and then decreases smoothly to the new stationary value 655–675 GtC over the course of approximately 2000 years (Fig. 7b). In this case, the maximum atmospheric carbon content reached over 65–75 years exceeds the current content by 80–100 GtC (11–13%), and the surface biota and the ocean absorb the excess of anthropogenic carbon even if the flux of anthropogenic emissions does not decrease with time. Over the course of almost 5000 years, the lithospheric reserves of fossil fuel run out, and the contents of carbon in the soil–vegetable system and the ocean stabilize at levels of 8880 and 40670 GtC, respectively, which exceed the current level by factors of 2.33 and 1.05 (Figs. 7c, 7e). The state of the carbon cycle is primarily determined by depletion of fossil fuel rather than a nonlinear character of interactions inside the system, and this is supported by the calculations of a steady state of the disturbed system from formulas (5)  $x^{(2)} = [653.21, 8842, 41665]$  GtC. The surface biota absorbs atmospheric carbon more intensively than the ocean, which is due to the neglect of saturation (for surface biomass) in the flux function  $f_{12}$ . Accordingly, the flux from soil and vegetation to the atmosphere, which corresponds to an autotrophic and heterotrophic respiration of vegetation and soil, also increases.

At a constant level of anthropogenic emissions, the sensitivity of the dynamics of reserves in the reservoirs to a change in the difference of the atmosphere–surface biota exchange fluxes can be seen in Fig. 8.

The upper curve corresponds to the coefficients of the flux  $f_{12}$ , which are calculated from the equilibrium and current schemes of the carbon cycle; the lower solid line is obtained from the values found with a refined method from (4); and the other two curves are obtained for the lower and upper boundaries of the measured difference  $f_{12} - f_{21}$ . It is seen that, with a cruder calibration of the coefficients of photosynthesis (the estimates of fluxes and reserves are taken into account at two times), the amplitude of the response of the atmospheric reserve to a constant anthropogenic emission is greater than that in the case when the estimates of the differences between opposite fluxes are used for three times. The maximum range of the response of the atmospheric carbon reserve under variations in the difference estimate for the 1980s (0.9 GtC) is almost 200 GtC (dashed curves), which suggests a rather high sensitivity of the response of the carbon cycle to anthropogenic disturbances to the coefficients of carbon absorption by the surface biota from the atmosphere.

The stability of atmosphere–oceanic steady states in which no carbon is present in surface ecosystems is determined by the form of the ocean–atmosphere interaction. With the calculated parameters of system (1), these theoretically possible equilibria are complex and do not occur.

#### OPERATION OF THE MODEL WITH A THIN ACTIVE LAYER OF THE OCEAN

According to the current notions [20], the mass of carbon in the upper quasi-homogeneous layer of the ocean is estimated by values on the order of 1000 GtC, and the fluxes characterizing the vertical exchange between the surface and the deep-water layers of the ocean are taken to be donor fluxes estimated at 40–42 GtC/year (Fig. 2b). The corresponding coefficients are given in Table 3. As in basic system (1), the preindustrial state is a stable equilibrium, but the relaxation times for this state are significantly shorter for the biota of the land and the ocean than for the entire oceanic reservoir involved in an active interaction with the atmosphere (Fig. 9). The undisturbed system returns to the preindustrial state, on average, over several decades. The carbon cycle with a continuous disturbance at a level of 7.2 GtC/year behaves in a different way: unlike the model with the entire ocean, the content of carbon in the upper layer decreases to 925 GtC over 2000 years, and, although the decrease is insignificant, we can say that the oceanic active layer becomes a donor of carbon transported through the atmosphere to the surface soil–vegetation complex. In this case, the dynamics of reserves in the atmosphere and surface biota is similar to that simulated for complete system (1) (Figs. 7a, 7c). Thus, the degree of involvement of oceanic carbon in the global cycle affects both the content of carbon in the water

depth and the direction of net exchange at the ocean–atmosphere boundary. A numerical experiment has shown that the selection of flux dependences in the pair  $\{f_{13}, f_{31}\}$  does not exert a significant effect on the dynamics of the content of carbon in major reservoirs. This dynamics remains unchanged if the types of con-

trol of these fluxes are changed so that the first of them becomes the Volterra type and the other becomes the donor type ( $f_{13} = \gamma_{13}x_1x_3$ ,  $f_{31} = \alpha_{31}x_3$ ). For this case, the equilibrium relations yield the steady states of the same type as in model (1). A complete nontrivial equilibrium has the form

$$\begin{aligned} x_1^{(2)} &= \frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}}, \\ x_3^{(2)} &= \frac{\alpha_{23}(\alpha_{41} + \alpha_{43})(M - x_1^{(2)})}{\alpha_{34}(\alpha_{23} + \alpha_{41}) + (\alpha_{41} + \alpha_{43})\left[\alpha_{23} + \alpha_{31} - \gamma_{13}\frac{L_{12}(\alpha_{23} + \alpha_{21})}{K_{12} - \alpha_{23} - \alpha_{21}}\right]}, \\ x_2^{(2)} &= \frac{1}{\alpha_{23} - \alpha_{43}}\left[\alpha_{43}\left(\frac{L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}} - M\right) + \left(\alpha_{31} + \alpha_{34} + \alpha_{43} - \frac{\gamma_{13}L_{12}(\alpha_{21} + \alpha_{23})}{K_{12} - \alpha_{21} - \alpha_{23}}\right)x_3^{(2)}\right], \end{aligned} \quad (7)$$

and its stability is determined by the roots of the characteristic equation of the Jacobi matrix:

$$J_M^{(2)} = \begin{bmatrix} a & b & c \\ d & 0 & 0 \\ f & g & h \end{bmatrix} = \begin{bmatrix} -\alpha_{41} - \alpha_{13} - \frac{K_{12}L_{12}x_2^{(2)}}{(L_{12} + x_1^{(2)})^2} - \gamma_{13}x_3^{(2)} & -\alpha_{41} - \frac{K_{12}x_1^{(2)}}{L_{12} + x_1^{(2)}} + \alpha_{21} & -\gamma_{13}x_1^{(2)} - \alpha_{41} + \alpha_{31} \\ \frac{K_{12}L_{12}x_2^{(2)}}{(L_{12} + x_1^{(2)})^2} & 0 & 0 \\ -\alpha_{43} + \gamma_{13}x_3^{(2)} & \alpha_{23} - \alpha_{13} & \gamma_{13}x_1^{(2)} - \alpha_{31} - \alpha_{34} - \alpha_{43} \end{bmatrix}, \quad (8)$$

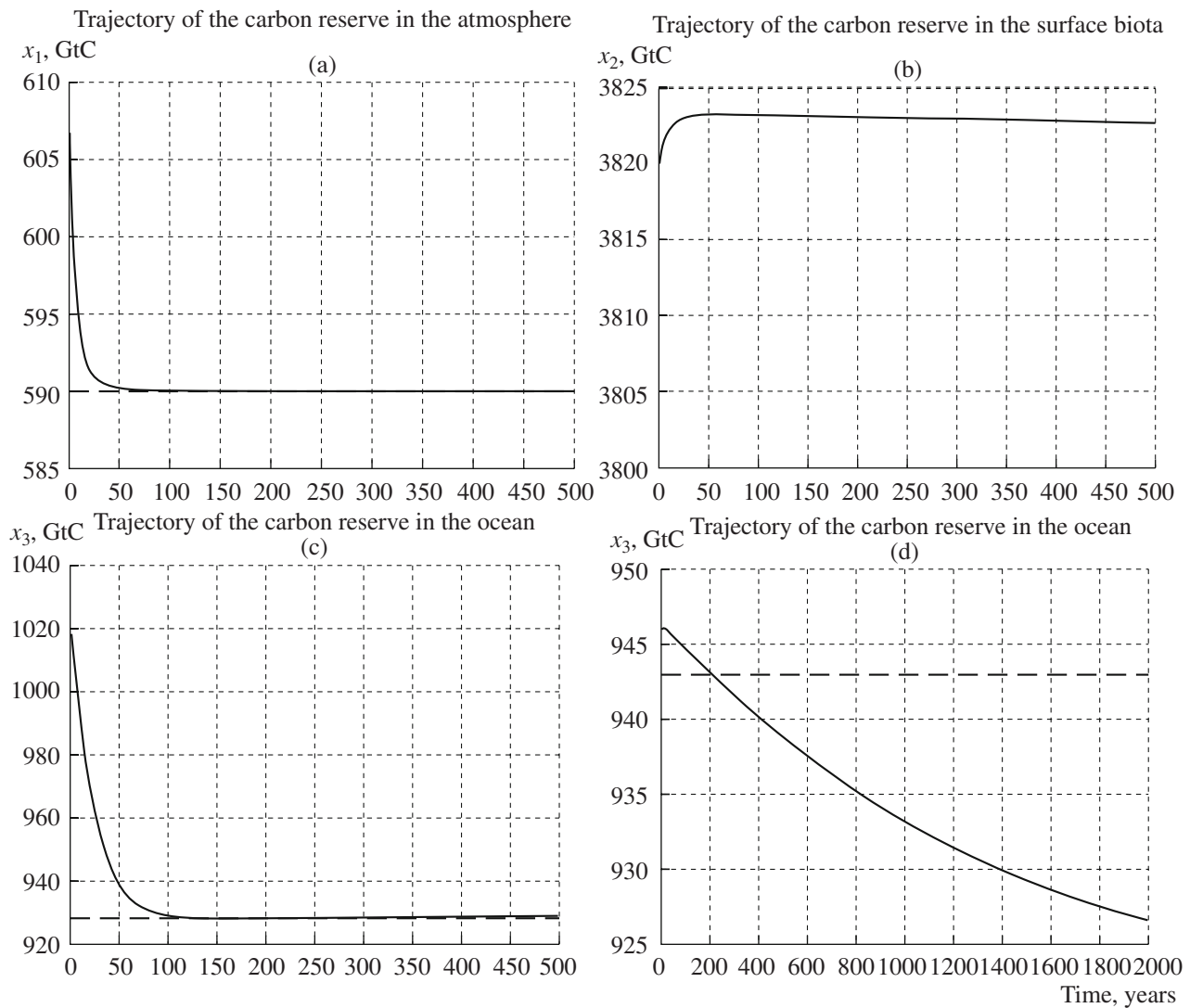
$\lambda^3 + k_2\lambda^2 + k_1\lambda + k_0 = 0$ , whose coefficients are expressed in terms of the matrix elements as in model (1). The coefficients of the characteristic equation yield the expressions for the stability boundaries according to the Routh–Hurwitz criterion. The multiplicity line  $k_0 = d(bh - cg) = 0$  of nontrivial equilibrium consists of two parts, one of which corresponds to the equality  $x_2^{(2)} = 0$  and lies outside the positive orthant of the parametric plane  $(\alpha_{41}, M)$ . Therefore, as in model (1), the preindustrial equilibrium is a stable node and remains such under an anthropogenic disturbance.

#### DYNAMICS OF THE CARBON CYCLE FOR SPECIFIED SCENARIOS OF ANTHROPOGENIC EMISSIONS

We are interested in simulation of the carbon cycle for the IPCC scenarios of emissions [15]. It is convenient to use such a scenario in models (1) and (1-M) with the aid of parameter  $q_{41}$ , which is varied at each time of integration in an interval from 1859 to 2100. Figure 10 shows the trajectories of variations in carbon reserves in the model reservoirs for scenarios SRES, A1B, A2, B1, and B2. The introduction of the flux corresponding to land use does not result in any

significant changes in the dynamics of these reserves. The values of carbon emissions in this process according to the same scenarios are introduced with the aid of the flux  $q_{21}$ , which supplements the flux of vegetation and soil respiration  $f_{21}$  at each time. Attention is drawn to the fact that, if the dynamics of carbon in the atmosphere varies from scenario to scenario (Figs. 9a, 9b), in all cases, for both models, the surface biota monotonically absorbs atmospheric carbon independently of the amount of its excess arrived from the lithosphere but only with different intensity (Figs. 10b, 11b). This is also due to the absence of explicit feedback between the content of carbon in the surface ecosystems and the flux of photosynthesis at  $\phi = x_2$  as well as at a constant level of emissions. The behavior of carbon in the ocean and the lake–river system is different for models (1) and (1-M) but similar for all scenarios. The first model demonstrates a decrease in oceanic carbon to 38650–38680 GtC for 2100 as compared to the current value 38728 GtC, and the other model shows its increase in the upper active layer up to 1050–1320 GtC as compared to the current value 928 GtC. In the first case, the absorption of carbon by growing surface vegetation exceeds the effect

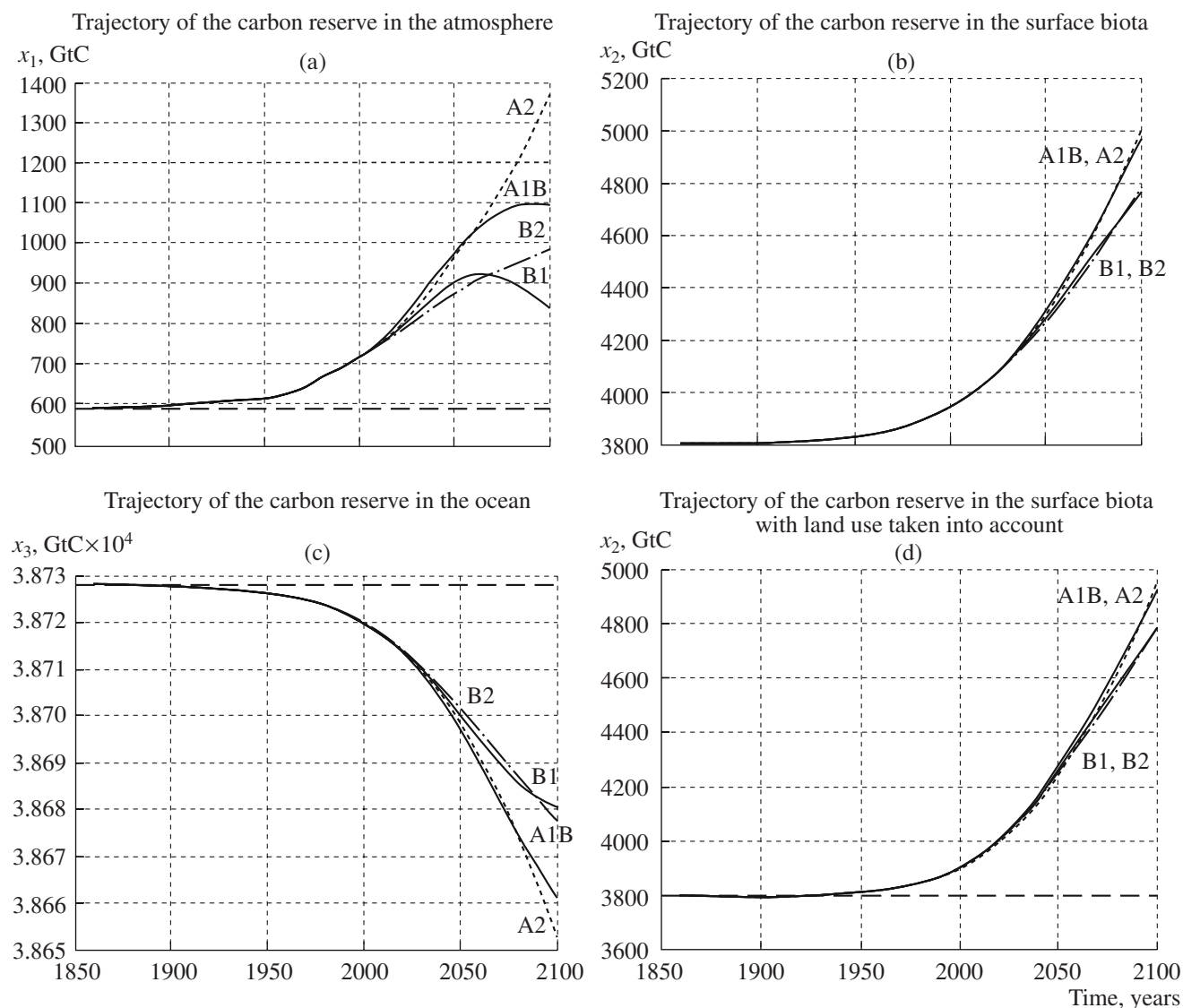




**Fig. 9.** Carbon reserve (GtC) in the (a) atmosphere, (b) surface biota, and (c) ocean over 500 years after a single disturbance of the preindustrial equilibrium in model (1-M) with the oceanic active layer. The dynamics of carbon in (d) the upper active layer at constant emissions at a level of the year 2000.

of the ocean, which becomes a weak source of  $\text{CO}_2$ , and, in the other case, the upper quasi-homogeneous layer, which serves as a sink for atmospheric carbon, absorbs its excess more intensively. Each of the versions can be explained by the corresponding model: the inertia of the entire ocean hinders it from a fast response to continuously increasing atmospheric emissions, and the surface biota has time to absorb more carbon, while the upper active layer can have a faster response. Comparison between the intensity coefficients for the fluxes  $f_{13}$  and  $f_{31}$  supports this conclusion. The dynamics of the content of carbon in the upper layer is different for different scenarios: for scenarios A2 and B2, its content increases monotonically to 1320 and 1130 GtC, and, for scenarios A1b and B1, its content increases to the maxima 1110 and 1170 GtC

over the period 2060–2080 and, then, by 2100, decreases to 1060 and 1155 GtC, respectively (Fig. 11c). The behavior of the atmospheric reservoir also depends on a concrete scenario in both models and has a qualitatively opposite character. Thus, a monotonic increase in the content of atmospheric carbon is characteristic of scenarios A2 and B2 in model (1) and scenarios A1B and B1 in model (1-M) with the upper active layer. In contrast, under scenarios A1B and B1 in the presence of the entire ocean, the maximum of atmospheric carbon reserve is reached and, after that, it decreases similarly to scenarios A2 and B2 in the case of the upper active layer (Figs. 10c, 11c). Thus, the proper trajectory of the flux of anthropogenic emissions, which is monotonic only for scenarios A2 and B2, does not always unambiguously cause a



**Fig. 10.** Short-term dynamics of the carbon reserve in major reservoirs for model (1) under different IPCC scenarios of anthropogenic emissions [15]: for the (a) atmosphere, (b) surface biota, (c) ocean, and (d) surface biota with land use taken into account. The emission scenarios are SRES, A1B, A2, B1, and B2.

monotonic increase in atmospheric carbon, wherein the nonlinear character of interaction between major reservoirs of the system is manifested and a specific effect of competition for atmospheric carbon between the surface biota and the ocean is reflected. Figures 10d and 11d show the response of the surface biota to the introduction of land use: in the first case, there is an almost monotonic increase in carbon reserves, and, in the second case, such an increase begins only from the present. The model shows that the increase of carbon in the surface biota exceeds the losses caused by its release during land use; however, to support or argue against this conclusion, it is necessary to pass on to a compartment scheme with separated soil and vegetation.

## CONCLUSIONS

The description of the global carbon cycle with the aid of a closed low-parameter compartment model yields an adequate qualitative and quantitative (under an accurate calibration) understanding of the operation of this cycle and its response to anthropogenic disturbances.

The basic compartment scheme of the model with four reservoirs can serve as a basis for the whole sequence of similar models, in which each subsequent model differs from the previous one by one reservoir. In this case, the totality of the fluxes characteristic of an additional reservoir adds some new details and properties to the dynamic pattern of the carbon cycle.

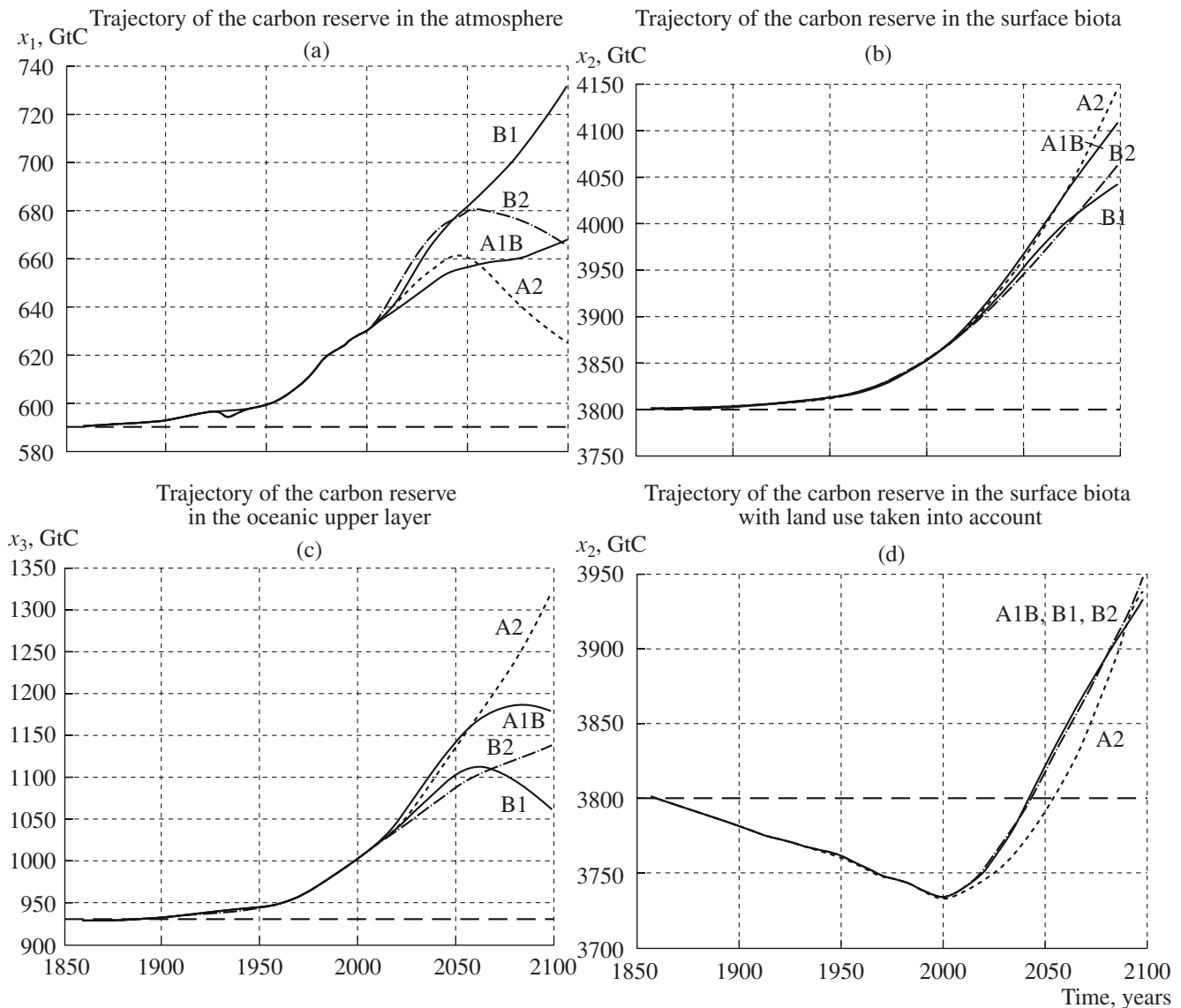


Fig. 11. Same as in Fig. 10 but for model (1-M) with the oceanic upper active layer.

Due to the features of their construction, low-parameter models give no way of describing the potentially possible dynamic regimes that differ from equilibrium or transient regimes.

In spite of the linearity of major interactions, the basic model with a simple photosynthesis saturation demonstrates a sufficiently simple behavior and allows one to estimate the times of relaxation of the Earth's carbon reservoirs for the preindustrial equilibrium or, conversely, to identify the coefficients of flux dependences according to the relaxation times known from paleoenvironmental data.

In the absence of anthropogenic carbon sources, equilibrium remains stable for small disturbances of both preindustrial and current states for the basic model of the carbon cycle and for the system with a small ocean. At a fixed current level of anthropogenic

emissions, the surface soil-vegetation reservoir absorbs the excess of atmospheric carbon after the latter reaches its maximum on the order of 850–870 GtC, stabilizes its atmospheric content at a level of 650 GtC, and increases by a factor of 2 to 2.5. Such an effect of surface-vegetation growth is probably related to the type of the flux function of photosynthesis, in which no self-limitation of the content of carbon in vegetation cover is present. Consideration for such self-limitation requires a change to a model of a larger (by one) dimensionality with separated vegetation and soil reservoirs.

The behavior of the ocean in the basic model and in the model with its active upper layer is noticeably different. The basic model demonstrates an increase in the carbon content in the entire reservoir, and, in contrast, the model with an active upper layer shows a

decrease in the active oceanic reservoir after a short-term (over several decades) increase in its absorption of atmospheric carbon. In the first case, the land and the ocean “eat away” atmospheric carbon, which serves as their resource, to the same extent (although with a different intensity). And, in the second case, the carbon absorbed by the land exceeds that absorbed by a comparatively thin upper layer of the ocean, and the land even encourages the transport of carbon from the oceanic active layer to the vegetation–soil reservoir. In the case of specifying continuous anthropogenic emissions, the potential equilibrium of the system goes beyond the limits of the permissible region, but no new dynamic regimes occur. The trajectories remain monotonic in both cases: when the emissions stop increasing and at their increase up to the exhaustion of fossil-fuel reserves. A more diverse behavior of the system is observed when the IPCC scenarios of anthropogenic emissions are used; in this case, the trajectory of emission values with time does not necessarily cause an increase in atmospheric carbon, which can decrease after its maximum is reached in different scenarios, depending on the degree of oceanic-carbon involvement in the carbon cycle.

#### ACKNOWLEDGMENTS

The authors are grateful to anonymous reviewers for helpful remarks.

This study was supported by the Russian Foundation for Basic Research, project no. 05-05-65137-a.

#### REFERENCES

1. B. Bolin, T. Degens, P. Duvigneaud, and S. Kempe, “The Global Biogeochemical Carbon Cycle,” in *The Global Carbon Cycle*, Ed. by B. Bolin, T. Degens, P. Ketner, and S. Kempe, (SCOPE, 1979), Vol. 13.
2. V. F. Krapivin, Yu. M. Svirezhev, and A. M. Tarko, *Mathematical Modeling of Global Biospheric Processes* (Nauka, Moscow, 1982) [in Russian].
3. K. Ya. Kondrat’ev and V. F. Krapivin, “Global Carbon Cycle and Climate,” *Issled. Zemli Kosmosa*, No. 1, 3–15 (2003).
4. A. M. Tarko, *Anthropogenic Changes of Global Biospheric Processes* (Fizmatlit, Moscow, 2005) [in Russian].
5. E. M. Volodin, “Atmosphere–Ocean General Circulation Model with the Carbon Cycle Model,” *Izv. Akad. Nauk, Fiz. Atmos. Okeana* **43**(3), 298–313 (2007) [*Izv., Atmos. Ocean. Phys.* **43**(3), 266–280 (2007)].
6. Y. M. Svirezhev, “Simple Spatially Distributed Model of the Global Carbon Cycle and Its Dynamic Properties,” *Ecol. Model.* **155**(1), 53–71 (2002).
7. V. A. Kostitsyn, *Evolution of the Atmosphere, Biosphere, and Climate* (Nauka, Moscow, 1984) [in Russian].
8. E. P. Borisenkov and Yu. A. Pichugin, “Multiyear Dynamics of the Components of a Climate System under the Conditions of Anthropogenic Action,” *Izv. Akad. Nauk, Ser. Geograf.*, No. 2, 27–38 (2004).
9. E. P. Borisenkov and Yu. A. Pichugin, “Assessment of the Consequences of Occurring Scenarios of Anthropogenic Loading on the Environment,” *Izv. Akad. Nauk, Ser. Geograf.*, No. 2, 7–17 (2006).
10. Y. M. Svirezhev and W. von Bloh, “Climate, Vegetation and Global Carbon Cycle: The Simplest Zero-Dimensional Model,” *Ecol. Model.* **101**(1), 79–97 (1997).
11. S. I. Bartsev, A. G. Degermendzhi, and D. V. Erokhin, “Global Minimum Model for Multiyear Dynamics of Carbon in the Biosphere,” *Dokl. Akad. Nauk* **401**(2), 233–237 (2005).
12. Y. M. Svirezhev and W. von Bloh, “A Zero-Dimensional Climate–Vegetation Model Containing Global Carbon and Hydrological Cycle,” *Ecol. Model.* **106**(1), 119–127 (1998).
13. O.-Y. Kwon and J. L. Schnoor, “Simple Carbon Models: The Atmosphere–Terrestrial Biosphere–Ocean Interaction,” *Global Biogeochem. Cycles*, No. 8, 295–305 (1994).
14. T. M. Lenton, “Land and Ocean Carbon Cycle Feedback Effects on Global Warming in a Simple Earth System Model,” *Tellus. B* **52**, 1159–1188 (2000).
15. *Climate Change 2001: The Scientific Basis Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change (IPCC)*, Ed. by J. T. Houghton et al. (Cambridge Univ. Press, Cambridge, 2002).
16. I. I. Mokhov, A. V. Eliseev, and A. A. Karpenko, “Sensitivity of the IAP RAS Global Climate Model with an Interactive Carbon Cycle to Anthropogenic Actions,” *Dokl. Akad. Nauk* **407**(3), 400–404 (2006).
17. A. V. Eliseev, I. I. Mokhov, and A. A. Karpenko, “Climate and Carbon Cycle Variations in the 20th and 21st Centuries in a Model of Intermediate Complexity,” *Izv. Akad. Nauk, Fiz. Atmos. Okeana* **43**(1), 3–17 (2007) [*Izv., Atmos. Ocean. Phys.* **43**(1), 1–14 (2007)].
18. S. M. Semenov, “Modeling an Anthropogenic Disturbance of the CO<sub>2</sub> Global Cycle,” *Dokl. Akad. Nauk* **398**(6), 810–814 (2004).
19. S. M. Semenov, *Greenhouse Gases and the Earth’s Present-Day Climate* (Meteorologiya i Gidrologiya, Moscow, 2004) [in Russian].
20. *The Global Carbon Cycle: Integrating Humans, Climate, and the Natural World*, Ed. by C. Field and M. Raupach (SCOPE, Island Press, Washington, 2004), Vol. 62.
21. K. L. Denman, G. Brasseur, A. Chidthaisong, et al., “Couplings between Changes in the Climate System and Biogeochemistry,” in *Climate Change 2007: The Physical Science Basis, Contribution of Working Group I to the 4th Assessment Report of the IPCC* (Cambridge Univ. Press, Cambridge, 2007).
22. B. Moore and B. H. Braswell, “The Lifetime of Excess Atmospheric Carbon Dioxide,” *Global Biogeochem. Cycles*, No. 8, 23–38 (1994).
23. D. O. Logofet, “Svicobians of the Compartment Models and DaD-Stability of the Svicobians: Aggregating “0-Dimensional” Models of Global Biogeochemical Cycles,” *Ecol. Model.* **104**, 39–49 (1997).
24. N. N. Zavalishin, “Dynamic Regimes of the Carbon Cycle Operation in an Ecosystem of Transition Wetlands,” *Mat. Model.* **17**(1), 43–64 (2005).
25. B. Adams, A. White, and T. M. Lenton, “An Analysis of Some Diverse Approaches to Modelling Terrestrial Primary Productivity,” *Ecol. Model.* **177**(3–4), 353–391 (2004).